

## Continuous Assessment Card

### Card No- 1. Biophysics and Biomolecules

| No. | Items                                                                                             | Marks(10 in each item) | Initials and date |
|-----|---------------------------------------------------------------------------------------------------|------------------------|-------------------|
| 1.  | Introduction of biochemistry, acid, base, $p^H$ , $p^K$ , buffer, Henderson-Hasselbalch equation. |                        |                   |
| 2.  | Solutions, crystalloid, colloid, dialysis and isotopes.                                           |                        |                   |
| 4.  | Carbohydrates.                                                                                    |                        |                   |
| 5.  | Lipids                                                                                            |                        |                   |
| 6.  | Amino Acids and Protein.                                                                          |                        |                   |
| 7.  | Enzymes, coenzymes, cofactors, isoenzymes                                                         |                        |                   |

### Card No- 2. Food, nutrition and vitamins

| No | Items                                                                                                                                             | Marks( 10 in each item) | Initial and date |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------|
| 1. | Basic concepts of Nutrient, food, diet, balanced diet, essential dietary components, , total calorie calculation,DRI, RDA, MR, BMR, BMI, SDA.     |                         |                  |
| 2. | Dietary fibers, nutritional importance of carbohydrate, lipid & protein, glycaemic index (GI) of food.                                            |                         |                  |
| 3. | Minerals- (macro & micro), trace elements, common nutritional disorders, PEM, BMI. obesity, iron metabolism and its deficiency, iodine deficiency |                         |                  |
| 4. | Water soluble vitamins                                                                                                                            |                         |                  |
| 5. | Fat soluble vitamins                                                                                                                              |                         |                  |

### Card No- 3. Digestion, absorption, bioenergetics and metabolism

| No | Items                                                                                                                                                                                      | Marks( 10 in each item) | Initial and date |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------|
| 1. | <b>Digestive juices</b> , local hormone of GIT, digestion & absorption of carbohydrate, lipid, protein.                                                                                    |                         |                  |
| 2. | <b>Bioenergetics</b> - biological oxidation, high energy phosphates, oxidative phosphorylation, respiratory chain. metabolism-definition, phases; anabolism, catabolism                    |                         |                  |
| 3. | <b>Carbohydrate metabolism</b> - glycolysis, fate of pyruvate, TCA cycle, HMP pathway, gluconeogenesis, glycogenesis, glycogenolysis, blood glucose regulation.                            |                         |                  |
| 4. | <b>Lipid metabolism</b> : lipolysis, Beta-oxidation of fatty acid, fate of Acetyl-CoA, ketone bodies, ketosis & its pathogenesis. Lipoproteins & their metabolism, Cholesterol metabolism. |                         |                  |
| 5. | <b>Protein metabolism</b> : Amino acid pool, Transamination, Deamination. Source & fate of ammonia, ammonia intoxication, Urea cycle.                                                      |                         |                  |

### Card No- 4. Renal biochemistry, body fluid, electrolytes and acid base balance

| No | Items                                                                                                                                                                          | Marks( 10 in each item) | Initial and date |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------|
| 1. | <b>Renal biochemistry</b> - GFR, tubular load, TM, renal threshold, plasma clearance, osmolar clearance, free water clearance, acidification of urine.                         |                         |                  |
| 2. | <b>Body fluid</b> - Body fluid compartments, daily water intake & output, water turnover, body fluid volume regulation, volume disorders and diuresis.                         |                         |                  |
| 3. | <b>Acid-Base Balance</b> - origin of acids & bases, maintenance of static blood pH. Acid base disorders, their compensation & correction, anion gap and base excess.           |                         |                  |
| 4. | <b>Serum Electrolytes</b> - Serum electrolytes & their reference ranges. Functions, regulations, hypo & hyper states of serum $[Na^+]$ , $[K^+]$ , $[Ca^{++}]$ & $[PO_4^{--}]$ |                         |                  |

### **Card No- 5. Clinical biochemistry and clinical endocrinology**

| <b>No</b> | <b>Items</b>                                                                                                                      | <b>Marks( 10 in each item)</b> | <b>Initial and date</b> |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|
| <b>1.</b> | Clinical biochemistry- S I unit, Laboratory hazards, Sample collection, Photometry. Clinical enzymology, lipid profiles of blood. |                                |                         |
| <b>2.</b> | Clinical enzymology and lipid profiles of blood.                                                                                  |                                |                         |
| <b>3.</b> | Diagnosis of diabetes mellitus. OGTT, IGT, IFG and HbA <sub>1c</sub> .                                                            |                                |                         |
| <b>4.</b> | Thyroid function tests and interpretation.                                                                                        |                                |                         |
| <b>5.</b> | Commonly done LFT. Jaundice.                                                                                                      |                                |                         |
| <b>6.</b> | Renal function tests and interpretation.                                                                                          |                                |                         |

### **Card No- 6. Fundamental of molecular biology and genetics**

| <b>No</b> | <b>Items</b>                                                                                                  | <b>Marks(10 in each item)</b> | <b>Initial and date</b> |
|-----------|---------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|
| 1.        | Nucleic acids, nucleotides, DNA, RNA, DNA organization, Cell cycle.                                           |                               |                         |
| 2.        | The central dogma, Genome, Gene, Genetic code, Codon, Mutation, mutagens, Genotype, Phenotype, trait, allele. |                               |                         |
| 3.        | Replication, Transcription and post transcriptional modification,                                             |                               |                         |
| 4.        | Translation and post translational modification.                                                              |                               |                         |
| 5.        | Recombinant DNA technology, PCR, Cloning.                                                                     |                               |                         |